

Here's your **Ultra-Compact Pandas Cheat Sheet** - only the methods you'll use 90% of the time in ML:

PANDAS CORE METHODS (Quick Reference)

1. Loading & Viewing Data

python

`pd.read_csv('file.csv')` # Load CSV

`pd.read_excel('file.xlsx')` # Load Excel

`df.head()` # First 5 rows

`df.tail()` # Last 5 rows

`df.sample(10)` # Random 10 rows

`df.info()` # Column types + null count

`df.describe()` # Stats (mean, std, min, max)

`df.shape` # (rows, columns)

`df.columns` # Column names

`df.dtypes` # Data types

2. Selecting Data

python

`df['col']` # Single column (Series)

`df[['col1', 'col2']]` # Multiple columns

`df.iloc[0:5, 0:3]` # By position (rows, cols)

`df.loc[0:5, ['col1']]` # By label/name

`df[df['age'] > 30]` # Filter rows

`df.query('age > 30')` # Filter with string

3. Handling Missing Values

python

`df.isnull().sum()` # Count nulls per column

`df.dropna()` # Drop rows with nulls

`df.dropna(axis=1)` # Drop columns with nulls

```
df.fillna(0)          # Fill nulls with 0
df.fillna(df.mean())  # Fill with mean
df.fillna(method='ffill') # Forward fill
```

4. Adding/Modifying Columns

```
python
df['new_col'] = values  # Add new column
df['new'] = df['a'] + df['b'] # Create from existing
df.rename(columns={'old':'new'}) # Rename columns
df.drop('col', axis=1)  # Delete column
df.dropna(inplace=True) # inplace = modify original
```

5. Grouping & Aggregating

```
python
df.groupby('col')['value'].mean()  # Group by + mean
df.groupby('col')['value'].agg(['mean', 'sum', 'count'])
df['col'].value_counts()           # Frequency count
df.corr()                         # Correlation matrix
```

6. Categorical Encoding (ML Prep)

```
python
pd.get_dummies(df, columns=['cat_col']) # One-Hot Encoding
df['col'].astype('category')            # Convert to category
df['col'].map({'A':1, 'B':2})           # Manual mapping
```

```
# Label Encoding
```

```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['encoded'] = le.fit_transform(df['cat_col'])
```

7. Sorting & Ordering

```
python
df.sort_values('col')    # Sort ascending
df.sort_values('col', ascending=False) # Descending
df.sort_index()         # Sort by index
```

8. Combining DataFrames

```
python
pd.concat([df1, df2])    # Stack vertically (rows)
pd.concat([df1, df2], axis=1) # Stack horizontally (cols)
pd.merge(df1, df2, on='key') # Join (like SQL JOIN)
```

9. Applying Functions

```
python
df['col'].apply(lambda x: x*2)    # Apply to column
df.apply(lambda x: x.max() - x.min()) # Apply to rows/cols
df['col'].map({'A':1, 'B':2})    # Map values
```

10. Useful Checks

```
python
df.duplicated().sum()    # Count duplicates
df.drop_duplicates()    # Remove duplicates
df.nunique()            # Count unique values per column
df['col'].unique()       # Unique values in column
df.isnull().any()       # Check if any nulls
df.memory_usage()       # Memory usage
```

TOP 10 METHODS YOU'LL USE DAILY

#	Method	Use Case
1	pd.read_csv()	Load dataset

#	Method	Use Case
2	<code>df.head()</code>	Quick peek at data
3	<code>df.info()</code>	Check data types + nulls
4	<code>df.describe()</code>	See statistics
5	<code>df.isnull().sum()</code>	Find missing values
6	<code>df.fillna()</code>	Handle missing data
7	<code>df.groupby().mean()</code>	Analyze patterns
8	<code>pd.get_dummies()</code>	Encode categorical
9	<code>df['col'].value_counts()</code>	Check distribution
10	<code>df.corr()</code>	Feature correlation

QUICK ML WORKFLOW WITH PANDAS

python

Step 1: Load

```
df = pd.read_csv('data.csv')
```

Step 2: Explore

```
df.head()
```

```
df.info()
```

Step 3: Clean

```
df.isnull().sum()
```

```
df.fillna(df.median(), inplace=True)
```

Step 4: Encode Categorical

```
df = pd.get_dummies(df, columns=['gender', 'city'])
```

Step 5: Separate X and y

```
X = df.drop('target', axis=1)
```

```
y = df['target']
```

Step 6: Ready for ML!

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y)
```

Quick Interview Short Answers

Question	Short Answer
What is Pandas?	Python library for data manipulation and analysis
DataFrame vs Series?	DataFrame = 2D table, Series = 1D column
loc vs iloc?	loc = labels/names, iloc = integer positions
How handle nulls?	Drop, fill with mean/median/mode, or forward fill
What is get_dummies?	One-hot encoding for categorical variables